

analysis tools help find out the impulse response, step response, pulse response and tone pulse response (in the case of band-pass filters).

With the frequency domain analyses you can calculate power effective gain, power insertion gain, voltage effective gain, voltage insertion gain, current effective gain, current insertion gain, input and output impedance, group delay, phase, and the return loss of the filter.

System requirements

Operating System. Windows 8, Windows 7, Windows Vista, Windows XP, Windows ME, Windows 2000 and Windows 98.

Screen resolution. At least 800×600 pixels. Recommended setting is a 1024×768 resolution or greater.

Stages of design

Basically, there are five phases or stages in the design and analysis of a passive filter using AADE Filter Design.

Engineering specifications. This is the first stage, in which you select the application you want to perform and specify the parameters. With the help of the computer, you verify (by trial and error) whether the given set of specifications is met by any of the available filter types.

Synthesis. Select the type filter response to be used (Butterworth, Chebyshev, etc). A circuit diagram and parts list is created by the software from the given engineering specifications.

Analysis. The synthesised circuit can be analysed in the frequency and time domains using the different methods provided.

Plotting. The program plots the selected analysis and, hence, the results can be evaluated.

Optimisation. Results will typically be high or low on bandwidth at the first attempt, as viewed on the analyser. You can adjust the parameters of the filter schematic and repeat it as required.

Getting started

RUN the *fdinstall.exe* file. This opens the set-up window. Go through the



Fig. 3: Enter parameters for creating a Chebyshev LPF

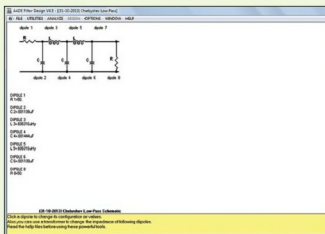


Fig. 4: A fifth-order Chebyshev LPF schematic created using AADE Filter Design

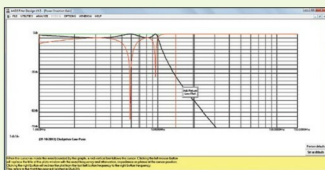


Fig. 5: Power insertion gain response of a fifth-order Chebyshev LPF

procedures and install the software. When the installation is complete, the program can be launched.

The main window would appear as shown in Fig. 1 with an 'Obligatory Commercial Message' of AADE's L/C Meter IIB. This message will appear for the first 10 times you open the program. At the bottom of the window, there is a region with a yellow background where helpful tips are dis-

played for almost all functions. You can turn it off by unchecking *Options* > *helpline*.

If some functions do not work properly after the installation, go to the menu bar: *Help* > *troubleshoot installation*. This opens a *Help* window with the necessary details to help you get to the bottom of the problem.

Design and analysis from the templates

The built-in templates of AADE are available in the menu bar under *Designs*, as shown in Fig. 2.

Here, as an example, let us build a low-pass Chebyshev filter.

First go to *Design* > *Chebyshev*. Select the function as *Low-pass* and click OK.

A window would appear as shown in Fig. 3. The following data must then be input.

F1 (-3dB cut-off frequency). The frequency which

should pass through the filter.

F2 (stopband frequency). The frequency above which the signal should block out.

Apass (passband ripple). Attenuation (in dB) at the cut-off frequency. Or in other words, how much of ripple is acceptable.

Astop (stopband attenuation). The minimum value of stopband attenuation (in dB). It is the amount of signal that